Evaluating Prediction Errors Nonparametrically and Parametrically

José Pedro Gama de Sousa Rodrigues

202006455 Inferential Statistics

INDEX

1. Introduction
2. Prediction Rules
3. Non-parametric Analysis
    a. Training and Test set
    b. Leave one out Cross Validation
    c. 10- Fold Cross-Validation
4. Parametric Analysis
    a. Mallows Cp Estimate
5. Conclusion
6. References

Introduction

Prediction is a crucial area in mathematics that enables us a further understanding in an area, leading to accurate forecasting about future events. To develop accurate predictions, mathematical models capable of understanding the distribution of observations are created. Since our course is Inferential Statistics, the following report is mainly about estimating the accuracy of those predictions developed through non-parametric (Cross-Validation, Train test set) and parametric statistical analysis (Mallows Cp).

The estimation of prediction accuracy is crucial for us to know if any predictions are appropriate and adequate with enough confidence to rely on. It allows us to compare models and detect if they are over or underfit.

To address these different approaches, the parametric and non-parametric analysis was conducted on the FPL dataset. FPL, is a game that prolongs throughout the football season, where each participant has a small budget to spend on a squad of 14 football players. This squad is then attributed points depending on the performance of each player every week, the points are added with a leaderboard of the participants being made, pitting each participant against each other. This FPL dataset contained 628 observations and 16 variables; each observation accounted for a football player in the English Premier league. 3 variables were categorical (Player, Team, Position) and the remaining 13 were numerical, mostly showcasing the player performance throughout the season.

The predictive models I decided to create used the number of Points a football player has at the end of a season as the dependent variable. This meant that for the prediction models, only 10 could be used as independent variables. Since only those 10 accounted for the player statistical performance throughout the season (Goals_scored, Assists, Saves, Yellow_Card, Red_card, Min_Played, CS – number of Clean sheets, BPS – Total sum of bonus points scored by player, Shots_on_target, Goals_conceded). The project was conducted on R studio.

Prediction Rules

Prediction rules *r(x)* are the models that develop predictors ($\hat{y}$) that will later be accessed for their accuracy through the parametric and non-parametric studies.

$$\hat{y} = r(x) \ \ for\ x \ \in \ X$$

The first rule developed in this project was multiple linear regression (MLR). MLR is a prediction model that uses more than one independent variable to predict y, this is done by creating a linear relationship between the explanatory and response variables. The assumptions of MLR include Homogeneity of variance, Independence of observations, Normality and of course linearity. MLR minimizes the sum of square residuals (RSS) creating the model that best fits the data. The MLR formula includes $\hat{y}$ as the predicted value, β_0 as the constant, β_i as the coefficients, X_i as the value of the independent variable at position i and the model error (ϵ):

$$\hat{y} = \ \beta_0 + \beta_1 X_1 + \cdots + \beta_n X_n + \ \epsilon$$

The Multiple Linear Regression model eventually used was:

$$\hat{y} = 1.376 + 4.55(Goals\ Scored) + 3.28(Assists) + 0.27\ (Saves) - 0.88(Yellow\ Card) - 2.79\ (Red\ Card) + 0.019\ (Minutes\ Played) + 1.6\ (CS) - 0.156\ (Shots\ on\ target) - 0.287\ (Goals\ Conceded) + 0.059\ (BPS)$$

Another rule I used to create the estimates was Ridge regression, also known as L2 Regularization. Ridge regression is also linear but has a methodology with the purpose of reducing prediction errors due to overfitting/ multicollinearity. This is done by adding a penalty (λ) to the calculation of the squared error. This leads to a shrinkage of coefficients towards 0 and a more robust prediction model. For the best results possible every variable should have the same weight meaning standardization is

recommended. The Ridge regression loss function is:

$$RSS_{Ridge} = \| Y - X\beta \|^2 + \lambda \| \beta \|^2$$

$$Penalty\ term\ (\lambda \| \beta \|^2) = \lambda \Sigma_{j=1}^{p} \beta_j^2$$

The First part of the function computes the residual sum of squares identical to the ordinary least squares regression stated above, while the second part is the regularization term that indicates the penalty. λ controls the extent to which the shrinkage to the coefficients is applied, this creates a bias-variance trade-off. A higher λ increases the penalty, reduces variability and increases bias. If λ is 0 the loss function becomes ordinary least squares regression.

I also did L1 regularization, or lasso regularization. Similarly to Ridge, it also has a λ penalty. But while ridge shrinks coefficients towards 0, the lasso model shrinks some coefficients to become exactly 0. Essentially being able to do feature selection. The Lasso loss function is:

$$RSS_{lasso} = \| Y - X\beta \|^2 + \lambda \| \beta \|_1$$

$$Penalty\ term\ (\lambda \| \beta \|_1) = \lambda \Sigma_{j=1}^{p} |\beta_j|$$

The standardization of the observations was explored but after a review of the results the variation of data explained was identical, so for a clearer comparison between models the raw dataset was used. The decision of the λ penalty was made after conducting a cross-validation of the regression models and a plot indicating the λ with the smallest CV error (lambda.min) and the largest λ value within one standard error of the minimum error (lambda.1se). For a more conservative approach I decided to make the analysis with lambda.1se.

$$\lambda_{Ridge} = 6.434233$$

$$\lambda_{Lasso} = 0.4233296$$

The Ridge regression Model with $\lambda = 6.434233$ had coefficients:

$$\hat{y} = 2.0198 + 2.946(Goals\ Scored) + 3.088(Assists) + 0.30\ (Saves) - 0.222(Yellow\ Card) - 1.632(Red\ Card) + 0.0078\ (Minutes\ Played) + 2.42\ (CS) - 0.544\ (Shots\ on\ target) - 0.160\ (Goals\ Conceded) + 0.056\ (BPS)$$

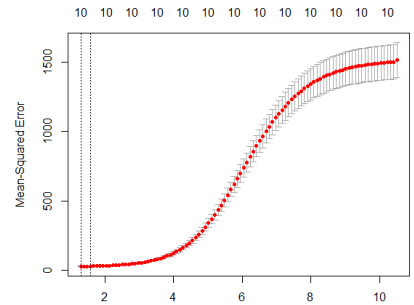

Fig.1. Plot of Cross-Validation of a Ridge regression. MSE in the y-axis and Log(λ) in the x- axis. Vertical lines indicate lambda.min and lambda.1se

The Lasso regression Model with $\lambda = 0.423329$ had coefficients:

$$\hat{y} = 1.641 + 4.06(Goals\ Scored) + 3.04(Assists) + 0.26\ (Saves) - 0.18(Yellow\ Card) - 0.54\ (Red\ Card) + 0.0\ (Minutes\ Played) + 2.33\ (CS) - 0.018\ (Shots\ on\ target) - 0.0\ (Goals\ Conceded) + 0.078\ (BPS)$$

For another perspective, a non-explicit model-based predictor rule was done for the non-parametric analysis. Regression trees, similar to decision trees, use nodes, branches and leaves but instead of attributing the value to a classification label it is able to identify continuous target values. It uses recursive partitioning to divide the data into smaller regions, eventually reaching a continuous value. It is used primarily to help easily visualize the decisions that lead to a specific outcome (Fig.3. Annex).

Non-parametric Analysis

Training and Test set

The first non-parametric evaluation method identified the accuracy of the prediction, through the division of the dataset into a 70:30 split of a training and test set. This method is the blueprint for all the cross-validation methods later analyzed. The training set $\boldsymbol{d_{train}}$ had a size of N = 440, while test set $\boldsymbol{d_{test}}$ was N_{test} = 188. Training set **d** was composed of N amount of pairs ($\mathbf{x_i}$,$\mathbf{y_i}$),

$\mathbf{x_i}$ being the vector of **p** predictors while $\mathbf{y_i}$ a real-valued response.

$$\boldsymbol{d_{train}} = \{(x_{i,}y_i), \qquad i = 1, \dots, 440\}$$

$$\boldsymbol{d_{test}} = \{(x_{0j}, y_{0j}), \qquad j = 1, \dots, 188\}$$

The prediction rule would then be fitted to the training set **d** and a model was created. This model would then be evaluated in three different ways:

The Root Mean squared error (RMSE), which quantifies the square-root of the average discrepancy between the predictions and the actual responses.

$$RMSE = \sqrt{\frac{1}{N}\Sigma_{i=1}^{N}(y_i - \hat{y}_i)^2}$$

The R^2 of a model, that showcases the proportion of variance in the dependent variable explained by the independent variables.

$$R^2 = 1 - \frac{Residual\ Sum\ of\ Squares}{Total\ sum\ of\ squares}$$

The Mean absolute error, which calculates the accuracy of the predictive model, through the average magnitude of the errors made by the predictive model.

Table 1. Train-Test evaluation results of the 4 Prediction rules

Model	RMSE	R^2	MAE
MLR	5.093	0.982	3.673
Ridge	6.532	0.974	4.421
Lasso	5.715	0.979	4.020
Tree	13.126	0.871	8.934

$$MAE = \frac{1}{N}\Sigma_{i=1}^{N}|y_i - \hat{y}_i|$$

The results showed that all models had a nice fit with the dataset, highlighting Linear regression and Regression trees as the best and worst fit, respectively.

Leave one-out Cross-validation

For LOOCV, the training set consisted of all the observations apart from one, the process was then repeated N times, providing an unbiased estimate of the RMSE, MAE and R^2. The LOOCV error function:

$$Err_{LOOCV} = \frac{1}{N}\Sigma_{i=1}^{N}D(y, \hat{y}_{(i)}), \quad \hat{y} = r_{d(i)}(x_i)$$

D stands for the loss function measuring the error between predicted and real target value.

Table 2. LOOCV evaluation results of the 4 Prediction rules

Model	RMSE	R^2	MAE
MLR	5.015	0.983	3.441
Ridge	5.861	0.979	4.099
Lasso	5.306	0.981	3.731
Tree	12.595	0.894	8.815

LOOCV is usually not suited for larger datasets, due to being computationally expensive and more susceptible to outliers. Since the number of observations in the FPL dataset was high with 628 observations, I suspected the evaluation metrics to be lower than their counterparts. But, contrary to my expectations, all the errors were lower than in the previous train test set. This could be due to a suboptimal split in the train test set.

K - fold Cross-Validation

K- fold CV is the most common Cross-validation method. The dataset is split into K groups of size N/K (in this case k = 10), where each group is used as the test for the training dataset composed by the remaining observations. The errors are then calculated after each group is fit and an average is made with the evaluation scores.

Table 3. 10 – fold CV evaluation results of the 4 prediction rules

Model	RMSE	R^2	MAE
MLR	5.002	0.984	3.475
Ridge	5.860	0.981	4.115
Lasso	5.194	0.983	3.713
Tree	13.015	0.896	9.068

As expected, the errors were generally lower in the regressor models. They also showed the same

comparison between models, MLR being the best fit.

Parametric Analysis

Mallows Cp Estimate

To calculate the Mallows Cp estimate we first need to understand covariance penalties. It starts with a set **d**, but while previously values were considered random, now they are fixed. In this calculation a new estimator $\hat{\boldsymbol{\mu}}$ is used as the predictor. $\boldsymbol{\mu}$ being the observed vector of responses y according to the probabilistic model. The new prediction rule can only be a regressor model making us only work with MLR, Ridge regression and lasso regression

$$\hat{\mu} = r(y)$$

We assume it to have a simple form where $\mathbf{y_i}$ is uncorrelated, an unknown $\boldsymbol{\mu_i}$ and a known variance $\boldsymbol{\sigma^2}$. $\boldsymbol{y} \sim (\boldsymbol{\mu}, \sigma^2, \boldsymbol{I})$. To Calculate the apparent error and prediction error the formulas are:

$$Apparent\ error(err_i) = (y_i - \hat{\mu}_i)^2$$

$$Prediction\ error\ (Err_i) = E_0\{(y_{0i} - \hat{\mu}_i)^2\}$$

Knowing the overall prediction error is the average of $\mathbf{Err_i}$. The covariance penalty lemma is then used to get the Mallows Cp formula. This lemma states that on average the apparent error underestimates $\mathbf{Err_i}$ by a covariance penalty 2cov($\hat{\boldsymbol{\mu}}_i, \boldsymbol{y}_i$).

$$E\{Err_i\} = E\{err_i\} + 2cov(\hat{\mu}_i, y_i)$$

The Mallows Cp estimate indicates whether a model is relatively unbiased regarding its model's coefficients and predictions. The Mallows Cp error estimate has the formula:

$$\widehat{Err}. = \frac{1}{N}\Sigma_{i=1}^{N}(y_i - \hat{\mu}_i)^2 + \frac{2}{N}\sigma^2 df$$

The first part showcases the residual sum of squares (RSS) and the second part the covariance penalty. The calculation of the degrees of freedom(**df**) for the prediction rule $\hat{\boldsymbol{\mu}}$**,** is done so that comparison between models can be made at a common ground, and has a general formula:

$$df = (\frac{1}{\sigma^2})\Sigma_{i=1}^{N}\widehat{cov}(\hat{\mu}_i, y_i)$$

Larger **dfs** are considered better fits to the models but need bigger covariance penalties. For MLR, the degrees of freedom equals to **p**(number of predictors). For lasso regression the **df** is equal to all the nonzero coefficients **p**. Finally, for Ridge regression if the penalty equals to 0, it acts the same as MLR(ordinary least squares) if not the degrees of freedom change depending on λ. Higher λ means lower degrees of freedom. A model is considered a good fit if the Mallows Cp value is close to **p**.

Table 4. Mallows Cp results and number of **p** predictors of the 3 regression rules

Model	Mallows C_p	p
MLR	24.01	10
Ridge	32.85	10
Lasso	26.88	8

Looking at the results we can see that the best fitted model is the multiple linear regression. Despite this, we can see that none of the models are near **p**. This suggests that the models may be overfit, and a more simplified model is more adequate.

Conclusion

In Conclusion, in all accuracy evaluation methods the multiple linear regression had the more accurate predictions and the best fitted models. The non-parametric error accuracy methods showed great results with low MSE values and almost 100% of variation explained. On the other hand, the parametric methods showed a completely different result showing us that models with a different number of predictors should be considered. For further research, a wider variety of models should be compared,

more parametric methods explored and a different study combining both parametric and non-parametric methods can be made, mallows Cp being used for variable decision and a cross-validation study after, with the new model, assessing the error.

References

1. Efron, Bradley and Hastie, Trevor. Computer Age Statistical Inference: Algorithms, Evidence and Data Science. https://hastie.su.domains/CASI_files/PDF/casi.pdf
2. Efron, The Estimation of Prediction Error: Covariance Penalties and Cross-Validation. https://people.eecs.berkeley.edu/~jordan/sail/readings/archive/efron_Cp.pdf
3. Lim, Buendia, Kim. The_role_of_prediction_in_the_teaching_and_learning_of_mathematics Lasso regression https://www.researchgate.net/publication/45164949_
4. Han, Kamber & Pei, Data Mining Concepts and Techniques
5. PenState University, Lesson 11.3 Best Subsets Regression, Adjusted R-Sq, Mallows Cp. https://online.stat.psu.edu/stat462/node/197/
6. Lin, Li. Practitioner's Guide to Data Science https://scientistcafe.com/ids/regression-model-performance#regression-model-performance

Annex

Fig.1. Plot of Cross-Validation of a Ridge regression. MSE in the y-axis and Log(λ) in the x- axis. Vertical lines indicate lambda.min and lambda.1se.

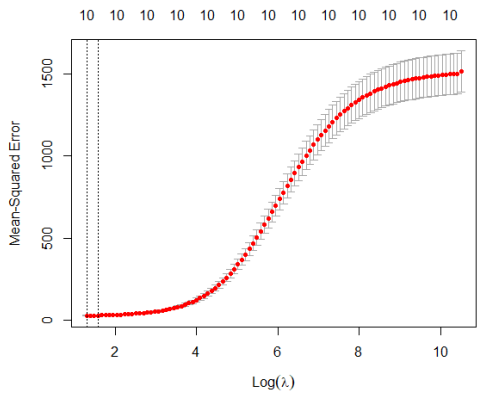

Fig.2. Plot of Cross-Validation of a Lasso regression. MSE in the y-axis and Log(λ) in the x- axis. Vertical lines indicate lambda.min and lambda.1se.

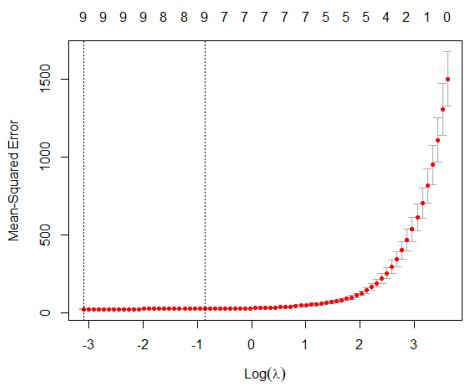
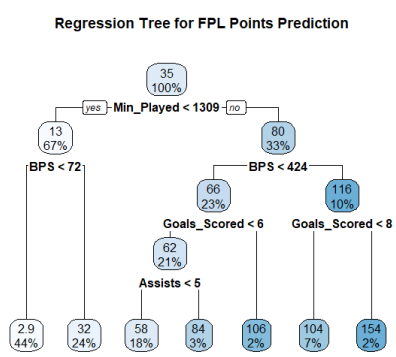
Fig.3. Regression Tree